MAKEATHON 2026 · TRACK 02 · NATURAL PRODUCTS

Terranova

Presentation & demo plan

The one-line version

Paracetamol relieves pain and destroys livers through two chemically separable routes. We screened **145,164** plant natural products for the first while filtering out the second - and then built the honest evaluation that tells us how far to trust the result.

What is in this document

Section	What it gives you
1. Read this first	A result that landed today and changes your framing
2. Why the order is the argument	What makes this structure work, and why it cannot be copied
3. The narrative	Six beats, timed, with what to say at each
4. The demo script	Exact click path through the dashboard
5. Numbers to know cold	The figures a judge may ask you to defend
6. Question prep	Eight likely questions with answers
7. Red lines	Claims you must not make
8. Pre-flight checklist	What to verify before you walk up

Status

All thirteen pipeline stages have now run. The three that were still missing this morning - external literature validation, the knowledge graph, and the written report - completed today, and the dashboard shows all of them.

1. Read this first

The result to lead with

The project has a genuine, externally validated result, and it is the only number here grounded in experiments nobody on this team ran. Lead with it.

Validation set	n	ROC-AUC	95% CI	p
CB1, all PubChem overlap	71 (53 act / 18 inact)	**0.952**	[0.89, 1.0]	5.3e-09
CB1, novel chemistry only	29 (13 act / 16 inact)	**0.779**	[0.533, 1.0]	0.0044

Why this one counts when the others did not

Every earlier validation compared against compounds somebody published as ACTIVE - no negatives, so no denominator. PubChem BioAssay records the outcome explicitly, Active and Inactive, so compounds that were screened and did nothing are in the data. Every one of these 71 compounds was excluded from all 44,516 training structures, so the model is predicting, not remembering. Median predicted score was 0.3444 for measured actives against 0.0001 for measured inactives.

The negatives are the persuasive part

It correctly REJECTED beta-caryophyllene and eleven of its isomers - the compound routinely called a "dietary cannabinoid" because it binds CB2, but which is not a CB1 ligand - scoring them around 1e-4. It also correctly rejected mescaline, DMT and O-methylbufotenine. A model merely pattern-matching "plant compound that sounds psychoactive" fails every one of those.

State the limits in the same breath

CB1 only. The TRPV1 overlap was 43 compounds all measured active and the FAAH overlap 13 all measured inactive - single-class, so ROC-AUC is undefined and we claim nothing for them. **Quote ROC-AUC, never enrichment:** PubChem skews active (74.6% base rate here) so enrichment factors are meaningless. **It is retrospective** - a blind test, since the model never saw these outcomes, but not a prospective one.

The second validation, which stayed inconclusive

A separate literature-based test ran earlier, and the first reading of it was **wrong in a way worth understanding**. Report it alongside the CB1 result rather than quoting only the number that worked.

The claim we nearly made, and why it was wrong

The raw output says: 16 held-out literature actives, none scoring above 0.5. Stated like that it sounds like total failure. Two things are wrong with it. **First the denominator.** Seven of those 16 compounds are weaker than the potency cut-off their own model was trained on - eugenol is reported at 300 uM against a 1 uM bar - so scoring them low is the correct answer, not a miss. **Second the bar.** A probability of 0.5 is not an activity boundary: measured against this library it sits near the 99th percentile, so "none above 0.5" really means "none reached the top 1% of 145,000 compounds".

What the corrected result actually says

Measure	Value
Held-out claims, raw	16

Measure	Value
Excluded: weaker than their model's own cut-off	7
Fair denominator	**9**
Median percentile against the 145k library	**55.1** (chance = 50)
Wilcoxon signed-rank vs chance	p = **0.3672**
Sign test	p = **0.5** (5 of 9 above chance)
Reaching the library's top 10%	1 of 9

The word is INCONCLUSIVE, not failed

We cannot reject the null and we cannot demonstrate signal either. Nine compounds was never going to settle the question. **Say "we have no evidence the screen generalises to plant chemistry", never "our model failed".** The first is what the data supports; the second is a claim you cannot defend and that a statistician in the room will take apart.

And we diagnosed it rather than shrugging

Two explanations were on the table. We tested both, which is the part most teams will not have done.

Explanation	Test	Result
Ruled out: our potency cut-off was too strict, so plant compounds were inactive by definition	Retrained TRPV1 and CB1 at the looser 10 uM bar, decoys matched	median 82.3 to 86.2 (+4.6). Top-10% hit rate unchanged at 3. It also made a false positive worse.
Confirmed: the training set never contained this chemistry	Six plant FAAH inhibitors, each scored by a model that saw the other five	**median 3.5e-05 to 0.330 - four orders of magnitude**

Why this is still the strongest thing you have

You ran an external validation, found it inconclusive, refused to spin it either way, then ran a second experiment to rule out the cheap explanation and a third to confirm the real one. That is a team debugging its own result in public. It is worth far more than a PR-AUC nobody can interrogate - and unlike a number, it cannot be produced in the last hour.

The honest three-line summary

1. Inside its training distribution the screen works: measured actives score a median 0.9988, measured inactives 0.0004.

2. On genuinely new plant chemistry we cannot yet say. Nine usable compounds, median percentile 55.1, p = 0.3672.

3. The limit is data coverage, not the model, the features or the threshold - and we measured each of those rather than assuming.

2. Why the order is the argument

The running order in the next section is not a stylistic choice, and changing it breaks the case. This page explains why, so that whoever is holding the clicker understands what each beat is doing.

The shape

It works, then here is where it breaks, then here is why, then here is the fix. Four steps, and your project has the evidence for all four.

Step	Your evidence	What it establishes
It works	Positive controls: 38 measured actives, median 0.9988; 47 measured inactives, median 0.0004	The machine is built correctly
Where it stops	External literature set: 9 usable compounds, median percentile 55.1 against 50 for chance, p = 0.3672	The limit of what we can claim
Why	FAAH rescue: median moves 3.5e-05 to 0.330	The mechanism of the failure
The fix	A few hundred curated literature actives; the rescue prices it	A roadmap, not an apology

Why every step is load-bearing

The clearest way to see it is to remove one and look at what is left.

If you present...	What a judge actually hears
"It works" alone	A demo. Unfalsifiable, and every team has one. It carries no information, because a team that never checked would show the identical slide.
"It breaks" alone	A failed project. Nobody presents this.
Works + breaks	Honest but defeatist. A limitation you cannot explain, and the last thing the room remembers is the failure.
Works + breaks + why	A diagnosis. This is the step that turns a bad number into a finding.
All four	**A research programme. The negative result becomes the justification for the next phase.**

> **The hinge is step three**
>
> This is not four nice things stacked up. Without the rescue experiment, an inconclusive test is simply a shrug. With it, the same result becomes evidence for a specific, testable claim about why - and the argument survives.

Why it cannot be copied in the last hour

Each step depends on infrastructure that had to exist before you knew you would need it. That is precisely what makes it credible.

Step	What it required, and when
It works	Leakage control designed into screen.py - removing training compounds and keeping them aside as controls. An architectural decision made when the screen was written. Retrofitting it means re-running the screen.
Where it breaks	resolve_literature.py: 24 compounds hand-curated from the primary literature, structures resolved through PubChem, potencies recorded with confidence levels. Hours of careful work, and not fabricable - a judge can check whether kaempferol really inhibits FAAH at 5 uM.

Step	What it required, and when
Why	experiment_faah_rescue.py: a leave-one-out design that retrains a model per compound. A designed experiment, not a metric computed from existing output.

> **It is expensive to fake, which is the whole point**
>
> A team that spent their time optimising a number cannot produce this at 11pm - not through laziness, but because what makes it possible is the evaluation infrastructure, and that had to be built alongside the model. The arc is not a presentation technique bolted on at the end. It is the visible output of having built the project a particular way.

Why judges reward it specifically

Consider the judge's actual problem. They see fifteen to thirty teams in a day, every one claims their model works, and they can verify none of it. So what they are really trying to estimate is: **does this team know whether their own thing works?**

A team reporting only successes provides no evidence either way. A team reporting a specific, quantified limitation has demonstrably run a test that could have embarrassed them, and did not flinch. In a format where nothing can be verified, that is the only credible signal available.

It also inoculates you completely. The strongest attack on any screening project is "this will not generalise". Once you have said it yourself, with a number, before anyone asks, there is no attack left. The worst thing a judge can say, you have already said, with more precision than they could.

Three rules for delivering it

#	Rule	Why
1	Order is non-negotiable: competence first, limitation second, never reversed	You earn the right to report the negative by first demonstrating the positive.
2	Beats 4 and 5 must be adjacent - no slide, no tangent, no pause for questions	If the room sits in the failure for even thirty seconds, the framing is lost.
3	Tone is finding, not confession	You went looking for this. Never "unfortunately our model did not work".

The transitions, where it is won or lost

(beat 3 to 4) *"That is the part that works. Now - here is what happened when we tested it properly."*

(beat 4 to 5) *"That number could have ended the project. It did not, because we could isolate why."*

(beat 5 to 6) *"Which means the fix is not a better model. It is about three hundred more rows of data."*

> **Land on that last line**
>
> It reframes everything behind it. You are not a team whose model underperformed. You are a team that knows exactly what their model costs to finish - and that is a different kind of team entirely.

3. The narrative - six beats

Timed for a six-minute slot. Cut beat 6 first if you are given five, and compress beat 2 if you are given four. The order matters: you earn the right to report a negative result by first showing the thing works.

0:00-0:45 The hook: one molecule, two pathways

Paracetamol is in every medicine cabinet on earth and we still argue about how it works. The current account is that it is a prodrug: the body converts it to AM404, which acts on TRPV1 and the cannabinoid system. Its liver toxicity runs through a completely different route - oxidation to NAPQI. Two pathways, chemically separable. That gap is the whole opportunity.

On screen: Open on the Hypothesis view. Point at the two arms.

0:45-1:30 What we built

We harvested every measured activity for those three targets from ChEMBL, curated 58,394 raw records down to 31,271 with every filter counted, trained models with a scaffold-grouped split, then screened 145,164 plant compounds from LOTUS - each one carrying a literature DOI back to the paper that found it in that plant.

On screen: Overview view. Let the funnel chart do the talking.

1:30-2:30 It works - here is the proof

Before trusting anything, we held out every library compound that appeared in training. Of those, the ones measured active score a median of 0.9988 - capsaicin, resiniferatoxin, THC, all recovered at the top. The ones measured inactive score 0.0004. The pipeline separates them cleanly, and the score distribution is sharply peaked near zero, so it is not simply calling everything active.

On screen: Screen explorer -> Positive controls tab. Scroll the top rows.

2:30-3:45 Validated against real experiments

Then we went looking for experimental data to test it against. PubChem BioAssay records inactive outcomes as well as active ones, which gives a real denominator. 71 of our library compounds have published outcomes and were never in our training set. ROC-AUC 0.952, p = 5e-09. Restrict it to novel chemistry and it is still 0.779. And it correctly rejected beta-caryophyllene - the compound everyone calls a dietary cannabinoid, which is not a CB1 ligand.

On screen: External validation view, the green card at the top.

3:45-4:30 And here is where it stops

Then we ran the test most screens skip: compounds reported active in the primary literature but absent from ChEMBL. And be careful here, because the first reading of this was wrong. Seven of the sixteen are weaker than the cut-off our own model was trained on - eugenol is 300 uM against a 1 uM bar - so scoring those low is correct. On the 9 we can fairly ask about, the median lands at the 55.1th percentile. Chance is 50. p = 0.3672. ***We cannot claim it works, and we cannot claim it failed.***

On screen: External validation view. Stop on the amber card. Let it land.

4:30-5:15 So we found out why

Two explanations. One: our potency cut-off was too strict. We retrained at the looser bar - it bought 4.6 percentile points, left the hit rate unchanged, and made a false positive worse. Ruled out. Two: the training set has never seen plant chemistry. Six plant FAAH inhibitors, each scored by a model that saw the other five - the prediction moves four orders of magnitude. The model is fine. The database is the gap.

On screen: The diagnosis slide: ruled out on the left, confirmed on the right.

5:15-6:00 What it is for, and what is next

What we hand over is not a discovery, it is a prioritised, filtered, provenance-tracked hypothesis list: 50 compounds, each traceable to a plant and a paper, each screened against the reactive chemistry that makes paracetamol dangerous. The knowledge graph makes every claim inspectable, and measured, literature and predicted edges look different on purpose. The next step is a few hundred curated literature actives to close the coverage gap we just measured.

On screen: Knowledge graph, then back to the shortlist.

4. The demo script

Start the dashboard before you are called up, not when you reach the podium. Have it already open on the Overview view with the browser at full screen.

Launch

In the project folder:

```
.venv\Scripts\python.exe -m app.run
```

Then `http://127.0.0.1:8000`. It opens the browser for you. Everything is precomputed - no model loads at request time, so nothing can stall mid-demo.

Click path

#	Where	What to do	What to say / point at
1	Overview	Land here. Do not scroll yet.	The four tiles: screened, in-domain, shortlist, curated records.
2	Overview	Scroll to the funnel.	145,380 down to 50, every stage counted.
3	The hypothesis	Click nav 02.	The two pathway arms. This is the whole idea in one picture.
4	Screen explorer	Click nav 07, then the Positive controls tab.	Capsaicin at 0.99999. The model finds what we already know.
5	Screen explorer	Type 'Zanthoxylum' in search.	Sichuan pepper sanshools - a real shortlist hit with traditional use.
6	Screen explorer	Click any row.	The compound modal: scores, safety flags, and the DOIs behind it.
7	External validation	Click nav 05. Pause on the red card.	This is the honest result. Let the silence work.
8	External validation	Scroll to the percentile chart.	Only the vanilloids surface. The pattern is informative.
9	Models	Click nav 04, scroll to FAAH rescue.	Four orders of magnitude. Coverage, not capability.
10	Models	Scroll to the DILI card.	Paracetamol scores 0.37 - low risk. The model is wrong, and we show it.
11	Validation	Click 'Open interactive graph'.	9,082 nodes. Drag one. Evidence classes are coloured differently.
12	ML decoder	Only if a judge is technical.	The glossary and the six claims with their evidence.

If the laptop dies

Everything the demo shows also exists as files: `results/REPORT.md` is the whole project in 16,000 words, `results/shortlist.csv` is the 50 candidates, and `results/knowledge_graph.html` opens in any browser with no server at all. Have that last one open in a second tab as insurance.

5. Numbers to know cold

If you can only memorise one column, memorise the right-hand one.

Scale

Quantity	Value
Raw ChEMBL activity records harvested	58,394
After curation (every filter counted)	31,271
Plant compounds in the LOTUS library	145,380
Screened after leakage control	145,164
Within applicability domain	114,969 (79.2%)
Flagged by a severity-3 structural alert	828
Flagged as PAINS (assay interference)	13,276
Final shortlist	50
Knowledge graph	9,082 nodes / 9,718 edges

Performance - and its ceiling

Measure	Value
Scaffold-split PR-AUC, across all models	0.879 - 0.983
Positive controls, measured active: median score	0.9988 (34 of 38 above 0.9)
Positive controls, measured inactive: median score	0.0004
PubChem CB1 validation, ROC-AUC	**0.952 on n=71, p=5.3e-09**
Same, novel chemistry only	**0.779 on n=29, p=0.0044**
Literature validation, fair denominator	**9 of 16 (7 below their model's cut-off)**
Their median library percentile	**55.1 vs 50 for chance**
Separable from chance?	**No - Wilcoxon p = 0.3672, sign test p = 0.5**
Threshold experiment (10 uM retrain)	+4.6 percentile, top-10% hit rate unchanged - ruled out
What 0.5 actually is, as a percentile	98.8 (TRPV1) / 99.8 (CB1) / 99.7 (FAAH)
FAAH rescue: median before / after	3.5e-05 -> 0.330
DILI model ROC-AUC (scaffold, 5 repeats)	0.889
DILI model on paracetamol	0.369 - i.e. low risk, and wrong

Design decisions you should be able to justify

Decision	One-sentence reason
Scaffold split, not random	ChEMBL is congeneric series; a random split leaks analogues and inflates the score.
60,000 decoys as presumed inactives	The measured active rate is 84% - publication bias, not biology.
Property-matched decoys	Unmatched ones let the model learn 'heavy and greasy = active' and transfer nothing.
1-10 uM band discarded	Forcing a label on genuinely ambiguous compounds teaches the model noise.
FAAH labelled at 10 uM as well as 1 uM	Plant FAAH inhibitors sit at 1-15 uM; a 1 uM cut trains the model to reject them.
Two safety layers, alerts weighted above the model	The alert set flags paracetamol at maximum severity; the DILI model calls it safe.
Evidence typed throughout the graph	A graph where a prediction looks like a measurement is a graph that misleads.

6. Question prep

1. Your scaffold-split PR-AUC is 0.98. Isn't that suspiciously high?

Yes, and we can tell you why. The decoys are easy negatives - random drug-like molecules - so the model is mostly separating 'real ligand' from 'arbitrary molecule', which is far easier than ranking among plausible candidates. That is exactly why we ran an external validation, and there we could not reproduce it. We treat the scaffold figure as an upper bound, not a result.

2. Has any of this been validated against real experiments?

Yes, for CB1. We took every library compound with a published PubChem BioAssay outcome that was absent from our training data - 71 compounds, 53 measured active and 18 measured inactive - and the model separates them at ROC-AUC 0.952, p = 5e-09. Restricted to chemistry unlike anything in training it is still 0.779, p = 0.004. That is a blind test on experiments we had no hand in. It is retrospective, and it is CB1 only - the other two overlaps were single-class and cannot be scored.

3. So your model doesn't work?

For CB1 it demonstrably does, against independent experimental outcomes. For TRPV1 and FAAH we have no such evidence and we do not claim any. It also works inside its training distribution - measured actives 0.9988, measured inactives 0.0004. On genuinely new plant chemistry our test came back inconclusive: 9 usable compounds, median percentile 55.1 against 50 for chance, p = 0.3672. That is too few to settle it. What we did establish is why it is hard, and what it would cost to find out.

4. Nine compounds is a tiny validation set. Isn't that the real problem?

It is, and we say so on the slide. It is also why we did not report the raw 'none above 0.5' number: with nine points that framing dresses up an underpowered test as a finding. A properly powered validation set is the second item on our next-steps list, immediately after the curated training actives.

5. What would you do with another week?

Curate several hundred literature actives for the three targets, the way we did the six in the rescue experiment, and retrain. The rescue tells us what that buys: four orders of magnitude on held-out plant chemistry. We would also add hard negatives rather than random decoys, so the scaffold number stops flattering us.

6. Why only LOTUS? What about COCONUT?

We checked. COCONUT is larger at around 695,000 structures, but its bulk downloads carry IDs, SMILES and computed scores - not the organism and per-compound DOI linkage. Getting that needs their 32 GB SQL dump. LOTUS gives referenced structure-organism pairs directly, and that provenance is what makes the knowledge graph defensible. Most of LOTUS is in COCONUT anyway via its Wikidata collection.

7. Isn't it circular that most of your shortlist comes from plants you picked for pain use?

Partly, yes. Seed-list compounds are 11.5% of the library and 72% of the shortlist, a 6.3-fold enrichment - but those plants are on the list because they have documented pain use, and their chemistry resembles the ChEMBL ligands. We treat it as a consistency check, not a discovery. The interesting rows are the 14 background-tier compounds that got there with no traditional-use prior.

8. Have you tested anything in a lab?

No, and we never claim otherwise. Every number is a computational prediction over compounds this project has not assayed. What we are handing over is a ranked, filtered, provenance-tracked hypothesis list for someone with a lab.

9. Why should I trust your safety filter?

Do not trust it on its own - that is why there are two layers. The DILI model scores paracetamol 0.37, i.e. low risk, because paracetamol and isoniazid are absent from its benchmark entirely. The structural alert set does flag paracetamol at maximum severity on the exact NAPQI mechanism. So we weighted the alerts above the model and documented why.

10. What is actually novel here?

Three things. The target panel is chosen from a specific mechanistic hypothesis rather than convenience. The evaluation is built to fail honestly - scaffold split, external validation, applicability domain, typed evidence. And the FAAH rescue experiment isolates data coverage from model capability, which is the question everyone waves at and nobody measures.

7. Red lines

This is a biomedical project presented to people who may know more biology than you do. One overclaim costs more credibility than every chart you built. These are not style preferences.

Never say	Say instead
"We found a new painkiller."	"We produced a ranked hypothesis list for lab testing."
"These compounds are safe."	"These compounds lack the structural features associated with paracetamol's toxicity."
"The model is 98% accurate."	"Scaffold-split PR-AUC is 0.98, but external validation shows it does not transfer."
"This compound inhibits FAAH."	"Our model predicts activity; it has not been assayed."
"Natural means safe."	"Plants make electrophiles on purpose - several of our own hits carry alerts."
"We validated the model."	"We ran an external validation and it exposed a coverage limit."

The disclaimer is on every page of the dashboard for a reason

If a judge asks whether any of this is ready for a human being, the answer is an immediate and unhedged no. Saying that quickly and plainly reads as competence, not weakness.

8. Pre-flight checklist

	Check	How
[ ]	Dashboard starts and all artefacts are present	Open http://127.0.0.1:8000 - no amber banner at the top means all files are there
[ ]	All eight views render in both themes	Click each nav item, then the theme toggle, and click through again
[ ]	The knowledge graph opens and is draggable	External validation -> Open interactive graph
[ ]	The compound modal shows DOIs	Screen explorer -> click any row -> scroll to Literature provenance
[ ]	Browser zoom set so tables are readable from the back	Ctrl+Plus two or three times; check the projector, not your screen
[ ]	Second tab open on results/knowledge_graph.html	Insurance if the server dies
[ ]	Everything committed and pushed	git status is clean; git push has finished
[ ]	You can say the word INCONCLUSIVE without flinching	Not 'it failed', not 'it works'. Practise the exact sentence.
[ ]	You know the fair denominator and why it is 9, not 16	7 compounds are weaker than their model's own cut-off.

> **Last thing**
>
> The temptation in the room will be to present the 0.98 and hope nobody asks. Do the opposite. Lead with the thing that works, then show the test that broke it, then show that you already know why. A team that can say 'here is where our model stops working, here is the measurement that proves it, and here is what it would cost to fix' is doing science. That is the wow factor, and it is the one nobody else can copy in the last hour.